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Lesson Notes

MODULE 5 | LESSON 4

MACHINE LEARNING EXTENSIONS OF RISK PARITY

Reading Time	30 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

In the last lesson, we were introduced to the concept of Risk Parity (RP) as an attempt to solve some of the well-known issues with the mean-variance optimization (MVO) portfolio and in comparison with the Kelly Criterion (KC) portfolio. We also learned about an extension of RP that reintroduces return forecasts into the optimization problem (thereby loosening certain constraints). In this lesson, we go further still by exploring yet another approach to RP, which relies on a machine learning (ML) technique known as k-medoids clustering. We see that even within the k-medoids clustering approach, there are still choices to make about how to select initial medoids, which can be tested with historical data using a variety of performance measures.

1. Risk Parity Meets Machine Learning

We have seen that risk parity has advantages and disadvantages in relation to mean-variance optimization (MVO) and the Kelly criterion portfolio. We have also seen various modifications of all of these approaches to portfolio construction that would seek to enhance them. The [first required reading](#) in this lesson by Duarte and de Castro falls squarely in the latter category, as it starts with the basic premise of RP (and its superiority over MVO) but builds on it with the help of clustering to avoid the instability associated with inverting a large covariance matrix. Clustering is, of course, a machine learning technique that you should be familiar with, but the discussion in this reading also provides you with the most important information about this algorithm. Since the Duarte and de Castro model relies heavily on the partitioning around medoids (PAM) algorithm, the [second "reading"](#) (actually a very helpful video) explains PAM again in the context of various k-means and k-medoid clustering algorithms.

This study serves us well on many levels because it:

- reviews the problems with MVO, such as instability.

- applies machine learning concepts and algorithms that you already know, e.g., partitioning, k -medoids clustering, naïve diversification. (Obviously, you should review Module 5 of Machine Learning if you want to review partitioning and clustering, if the descriptions and discussions in this reading are not clear.)
- extends and improves on the RP approach (and relaxed RP approach) by introducing new algorithms and methods, e.g., fuzzy clustering, distance matrix, performance proportional allocation.
- demonstrates a comparison of strategies using very relevant performance measures, i.e. annualized return, annualized volatility, maximum drawdown, turnover ratio (an important indicator for transaction costs), Sharpe ratio.

The asset allocation section (III.A.) should be very familiar to you from previous modules in this course. It consists of a short review of MVO in relation to Sharpe and the efficient frontier.

By the same token, III.B. should be somewhat familiar to you from your Machine Learning course, but even if not, this reading does a good job of describing clustering in general and k -medoids clustering specifically. Pay close attention to the steps in each of the algorithms. Try to understand the purpose of each step and the logic of the order of the steps.

The following is a list of methods and algorithms for which you should understand the logic and steps involved. This is not meant to be an exhaustive list of topics you are responsible for learning, but rather a representative list that indicates the kind of techniques you will want to know well—and for which sufficient study of the readings should prepare you.

- Hierarchical vs. partitional clustering
- k -medoids clustering algorithm (including initialization approaches)
- Partitioning Around Medoids (PAM) algorithm
- Fuzzy clustering
- Performance proportional allocation
- Balanced Capital Allocation

The following sections introduce or provide a bit more detail about concepts that come up in the reading.

2. Condition Number

The condition number relates to the instability that we associate with mean-variance optimization (MVO). Here, when we refer to the condition number, we mean specifically the condition number for inversion, as the condition number of a matrix is specific to a given problem or task related to that matrix (Moler). In this case, the condition number of a matrix is how sensitive its inverse is to perturbations in the original matrix. A high condition number (or a high “conditioning number”) indicates that a small error in the original matrix will result in a very inaccurate inverse matrix. The condition number is always 1 or greater. It can be calculated using the singular values of a matrix—that is, the elements of the diagonal matrix resulting from singular value decomposition (denoted by the lowercase sigma in the figure below). A more insightful interpretation, especially for our purposes, is that the condition number quantifies the effect of a small change in the input data, which the lemma below formalizes.

Figure 1: Condition Number Calculation and Lemma

The singular values define a **condition number** of a matrix as follows:

$$\kappa(A) := \frac{\sigma_n(A)}{\sigma_1(A)} = \frac{\|A\|}{\|A^{-1}\|^{-1}}$$

Lemma 1. If $Ax = b$ and $A(x + \delta x) = b + \delta b$ then

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\delta b\|}{\|b\|}$$

Source: Spielman, Daniel. "Lecture 2." *The Behavior of Algorithms in Practice*. 2002.
<https://www.cs.yale.edu/homes/spielman/BAP/lect2.pdf>.

To understand the potential significance of the condition number to the portfolio optimization problem, just recall the inverted covariance matrix in both the numerator and denominator of the equation for portfolio weights in a mean-variance optimization:

$$w_{MVO} = \frac{\Sigma^{-1}\tilde{r}}{\tilde{r}^T \Sigma^{-1} \tilde{r}}.$$

3. The Distance Matrix

The **distance matrix** is just a matrix, the elements of which are equal to the squared distance between the pairs of objects indicated by the row and column, as in the figure below. Just as a correlation matrix has ones along the diagonal, a distance matrix has zeros along the diagonal. Note that in the article by Duarte and de Castro, the authors use the distances themselves to populate the distance matrix, instead of the standard *squared* distance.

Figure 2: The Distance Matrix

$$A = \begin{bmatrix} 0 & d_{12}^2 & d_{13}^2 & \dots & d_{1n}^2 \\ d_{21}^2 & 0 & d_{23}^2 & \dots & d_{2n}^2 \\ d_{31}^2 & d_{32}^2 & 0 & \dots & d_{3n}^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_{n1}^2 & d_{n2}^2 & d_{n3}^2 & \dots & 0 \end{bmatrix}$$

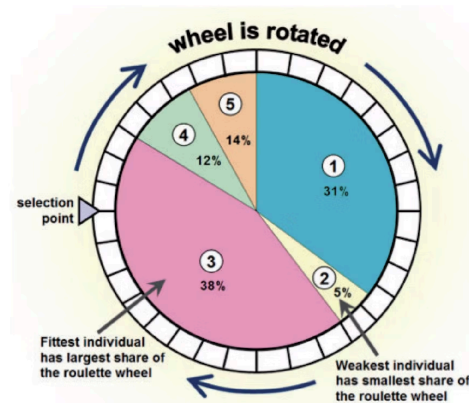
Source: "Euclidean distance matrix." *Wikipedia*. https://en.wikipedia.org/wiki/Euclidean_distance_matrix.

4. Performance Proportional Allocation

Duarte and de Castro borrow a technique from genetic programming in their formulation of a novel technique to construct a portfolio based on RP principles. They call this technique **performance proportional allocation** (PPA), and they use it twice, that is, at the levels of both the intergroup and the intragroup allocation. The genetic algorithm that inspired PPA is called **roulette wheel selection** (Duarte and de Castro 110780). Such a selection process is also known as fitness proportionate selection, and it might also serve as a metaphor for financial engineering itself, which seeks to recombine whatever techniques from whatever disciplines so long as they are "fit" for the purpose at

hand. The fitter the factor, the higher the probability that it will be selected by the algorithm for reproduction, as in the figure below. For a fuller definition, see the [third required reading](#), where this helpful graphic was adapted from.

Figure 3: Roulette Wheel Selection



Source: "Roulette wheel selection." *Computer Science Wiki*, 13 Jan 2022.
https://computersciencewiki.org/index.php/Roulette_wheel_selection.

Friendly reminder: If you have questions about any of this (or anything else!), use the forum to get your questions answered by the instructors or your fellow students. It is helpful to you to formulate what you do not understand—sometimes this process alone dispels the confusion. If you do understand something that one of your classmates does not, you will not only help your classmate by responding with your insight but you will also strengthen your understanding as you put it into words for someone else's benefit.

5. Conclusion

In this module, we discussed the the Kelly criterion (KC) and risk parity (RP) in their traditional forms in comparison to mean-variance optimization (MVO). We also investigated certain modifications of these portfolio construction techniques that sought to preserve their relative advantages (lower volatility, lower maximum drawdown, higher diversification, etc.) while improving on their weaknesses (often in terms of higher returns). We saw that these models in their original form were somewhat basic but also intuitively straightforward. Then, we reviewed potential enhancements to these models, which introduced sophistication and complexity as solutions to the original models' drawbacks.

In the next module, we continue to explore portfolio construction techniques, such as Factor Models, which take into account increasing amounts of information from the market. We also continue to apply state-of-the-art machine learning to enhance these models in terms of performance, which should by now be understood to mean both lower risk and higher return.

References

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